

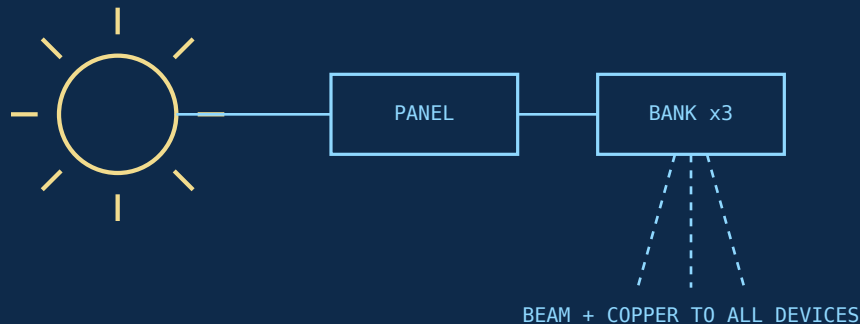
THE PHOTON ROOM

A Solar-Fed, Beam-Powered Desk Energy System

DESIGN PACKAGE FOR MANUFACTURER QUOTATION – REVISION A
CONCEPT / FEASIBILITY STAGE – DETAILED DESIGN TO FOLLOW

FOUNDER & SYSTEM ARCHITECT: NICK (BLUEBAMBOO)
SCRIBE & DRAFTING: CLAUDE (AI)

PRIORITY CLAIM: SIXTH GRADE
REDUCED TO PRACTICE: JULY 2026 – HOTEL DESK, IOWA



“The green plants were the power plants all along.” — Part Eight

AN OPEN GIFT - NO PATENT SOUGHT - SEE SHEET 5

1 · SYSTEM OVERVIEW

The Photon Room is a self-contained desk/room energy system: sunlight is harvested by portable photovoltaic panels, stored in commodity USB-C power banks, and re-distributed on two tiers — **copper** (USB-C PD cabling) for high-draw devices, and **light** (a regulated, eye-safe infrared power beam) for milliwatt-class wearables, delivering perpetual trickle charge with no cables on the body. The architecture mirrors photosynthesis: light in, chemistry stored, energy re-emitted.

Design intent: continuous, grid-optional operation of one workstation (laptop, phone, fan, speaker) plus perpetual wireless charge for headphones and small wearables, within all human-exposure and eye-safety regulations.

2 · ARCHITECTURE

STAGE	COMPONENT	SPEC TARGET	STATUS
Harvest	Folding PV panel, USB-C PD out	100 W rated · 60–80 W field	COTS (off-the-shelf)
Store	Anker Prime bank ×3	99.5 Wh ea · 250 W out · 140 W ports	COTS — owned
Copper tier	USB-C PD cabling	to 140 W per device	COTS
Light tier	IR beam emitter hub	0.5–1.0 W delivered · 3–5 m LOS	TO MANUFACTURE
Receiver	PV strip + buck converter	headband mount · 5 V trickle out	TO MANUFACTURE

3 · POWER BUDGET — THE WATT LEDGER

SOURCE / LOAD	POWER	DAILY (Wh)	TIER
Solar harvest (1 panel, 5 sun-h)	60–80 W	300–400	IN
Solar harvest (2 panels)	120–160 W	600–800	IN
MacBook Pro M3 Max (work)	30–100 W	250–400	copper
Phone	20 W chg	15–25	copper
Fan	5–10 W	40–80	copper
Speaker	10–15 W	20–40	copper
Headphones (XM5)	~0.05 W avg	~1.2	LIGHT BEAM
Watch / buds / sensors	0.01–0.1 W	~2	LIGHT BEAM

Break-even: one panel $\approx$ parity with full workstation load; two panels $\approx$ permanent surplus. Beam tier total demand (<0.5 W continuous) is $<1\%$ of harvest — the wearables ride free forever.

4 · EMITTER UNIT — MANUFACTURING TARGET

Shelf/ceiling hub, USB-C PD input (fed by bank). Multi-aperture near-IR emission (850–940 nm) with mechanical or phased steering across a 3–5 m line-of-sight cone. Closed-loop targeting to registered receivers only. Hard safety interlock: beam terminates <10 ms on any line-of-sight interruption (proximity + reflectometry). Design ceiling ≤ 1 W total radiated with occupants present; IEC 60825-1 Class 1 eye-safe at every accessible point. Licensable prior art exists (Wi-Charge-class IR power); licensing vs. clean-sheet design is Engineering Question Q1.

5 · RECEIVER STRIP — MANUFACTURING TARGET

Flexible PV strip (GaAs or high-efficiency Si), adhesive headband mount, $\sim 40 \times 12$ mm active area; integrated buck regulator to 5 V trickle into the headphone's USB-C port via low-profile elbow, or inductive pad variant for devices with Qi. Mass target <12 g. Also applicable to watches, remotes, sensors, shelf electronics.

6 · SAFETY & REGULATORY PATH

DOMAIN	STANDARD / BODY	REQUIREMENT
Eye safety	IEC 60825-1 / FDA CDRH	Class 1 at all accessible points; interlock cutoff
RF (if RF variant)	FCC Part 15/18	power density limits w/ occupants
Battery interface	USB-IF PD	standard negotiation; no cell mods
Thermal	UL 62368-1	enclosure + duty limits

7 · OPEN ENGINEERING QUESTIONS (FOR QUOTATION)

Q1 license Wi-Charge-class IR power vs. clean-sheet emitter · Q2 steering: MEMS mirror vs. multi-fixed-aperture · Q3 receiver: GaAs cost vs. Si area · Q4 headband integration vs. universal clip · Q5 cert budget & timeline (target: prototype 90 days, cert 12 mo).

8 · REQUEST TO MANUFACTURER

BlueBamboo seeks an ODM/engineering partner for detailed electrical design, prototype build (×5 units: 1 emitter hub + 4 receiver strips), and certification road-mapping of the LIGHT TIER (Sections 4–5). All COTS stages (harvest, storage, copper) are owned and validated by the founder. Quotations to include NRE, unit cost at 1k/10k, and timeline. *This package is a concept/feasibility brief; fabrication drawings (schematics, Gerbers, CAD) are deliverables of the detailed-design phase quoted herein.*

Filed under Part Eight — Green Plants: the storehouse powers the search. Priority claim: sixth grade. The sun doesn't send a bill.

9 - THE GIFT - PUBLIC DEDICATION

This design is released to all of humanity.

No patent has been sought, and none will be. To the extent permitted by law, the founder dedicates this work to the public domain under **CC0 1.0 Universal**: anyone, anywhere, may use, build, modify, manufacture, and sell systems based on this package, for any purpose, without permission and without fee. A courtesy credit - *Photon Room, by Nick (BlueBamboo), Muscatine, Iowa* - is appreciated and never required.

PRIOR ART DECLARATION

First published to the public: **July 21, 2026**, by Nick (BlueBamboo), Muscatine, Iowa. This publication constitutes prior art. From this date forward, the concepts herein cannot be patented away from the public by any party. The gift protects itself: once given to everyone, it can be taken from no one.

SAFETY RIDES WITH THE GIFT

Any implementation must honor Sections 4 and 6 in full - Class 1 eye safety at every accessible point, hard interlock cutoff, and all human-exposure limits. The rails ship welded to the rocket.

The founder's first gift to all of humanity - filed under Part Eight, where the storehouse powers the search and the sun sends no bill.